

Injection Moulding - Cushion Stability and Monitoring

1. What the cushion is [p1]

The cushion is the residual melt left in front of the screw at the end of injection and hold, nominally 3 to 10 mm. It guarantees the screw never bottoms out and that holding pressure is transmitted into the cavity. A stable cushion is a precondition for repeatable parts.

2. The instability threshold

Shot-to-shot cushion variation greater than 0.5 mm, or a cushion that steadily falls over a run, signals that melt is leaking backward during injection. This is the primary measurable sign used to flag a non-return-valve problem.

3. Causes of cushion instability

In order of likelihood: a worn non-return valve / check ring (most common); a mis-set decompression (suck-back) stroke; plasticising screw and barrel wear; inconsistent feeding; and resin viscosity or moisture variation. Rule out the setting and the resin before mechanical replacement.

4. Monitoring (SPC)

Trend the cushion on an SPC chart and alarm when the shot-to-shot range exceeds 0.5 mm or a downward trend appears. Pair cushion with recovery time and the injection-pressure integral for a complete picture of plasticising health.

5. Action

If the decompression setting is correct and the resin is in spec but the cushion still drifts, inspect and replace the non-return valve assembly (see the screw-tip-assembly reference).